

Adam MOUNIR

AI Research Student

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EDUCATION

EFREI - Paris Panthéon-Assas

MEng. - *Diplôme d'Ingénieur in Data & Artificial Intelligence*

• **Courses :** Graph Theory - Convex Optimization - Machine / Deep Learning - Reinforcement Learning - Natural Language Processing - Statistics/Probability - Chatbots & Generative AI - Big Data Ecosystems - MLOps - Cloud Computing

ENS Paris-Saclay - MVA - *Deep Learning in Practice (G. Charpiat)* - **Guest Auditor**

• **Key Topics:** Optimization Landscapes, Symmetry & Invariance, Generative Diffusion Models, Implicit Bias

University of California, Irvine

Exchange Student, MEng. in Computer Science

• **Courses :** Mathematics for Cryptography - Java Programming - Computer Architecture - Network Protocols

Paris, FR

2021 - 2026

California, USA

2023 - 2024

PROFESSIONAL EXPERIENCE

Inria - TAU (Université Paris-Saclay)

AI Research - Intern

Paris, FR

Apr 2026 - Sep 2026

Researching deep-learning architectures for end-to-end electroencephalography (EEG) decoding for the open-source library Braindecode

- Design transformer, graph-neural-network, convolutional and Riemannian models for neural decoding using open EEG datasets
- Co-organize the NeurIPS 2026 EEG decoding competition (Meta, top academic labs) building its benchmark infrastructure
- Supervised by Inria TAU (Sylvain CHEVALLIER, Guillaume CHARPIAT), Yneuro (Bruno ARISTIMUNHA, Arnault CAILLET)

Thales

AI Research - Intern

Paris, FR

Nov 2024 - Apr 2025

Researched global-scale Root Cause Analysis by distilling multi-step reasoning capabilities from LLMs into specialized architectures

- Designed a supervised alignment pipeline to map expert causal trajectories from logs into internalized reasoning schemas
- Distilled causal traces via GPT-4o to fine-tune a SFT pipeline using Llama3.1 (QLoRA) targeting latent reasoning convergence
- Achieved a 0.64 F1-score in RCA prediction and established a validation protocol for Out-of-Distribution scenarios

Société Générale

AI Engineer - Apprentice

Paris, FR

Oct 2025 - Apr 2026

Detected semantic discrepancies between unstructured documents disclosures and structured multi-provider data

- Automated cross-source data alignment and causal influence tracing within hierarchical scoring models
- Engineered a Zero-shot NER pipeline using Claude 3.5 Sonnet (RAG) and a Directed Graphs for topological impact analysis

RESEARCH PROJECTS

Braindecode

Open-Source Contribution | [Github Link](#)

Maintaining and extending the reference PyTorch/skorch library for end-to-end neural time-series decoding

- Implementing transformer, graph-neural-network and Riemannian architectures for EEG/MEG signal decoding
- Building reproducible benchmarking pipelines over open neurophysiology datasets (MOABB integration)
- Contributing tested and documented modules to a community-driven library used across academic and clinical labs

Nullflow : Neuro-AI for Complementary Learning Systems

Personal Research Project | [Preprint Link](#)

Tackling catastrophic forgetting in Deep Neural Networks during non-stationary sequential data streams

- Designing bio-inspired architectures optimizing the stability-plasticity trade-off via CLS Theory & Null-Space Projection
- Built a Wake-Sleep/Flow Matching framework in PyTorch for NREM/REM latent generative replay (~2.1M parameters)
- Outperformed 8 CL baselines by +18.8pp via 6-step ODE latent replay;

ACHIEVEMENTS

Google Deepmind x {Tech : Europe}

AI Hackathon - Big Berlin Hack

Berlin, DE

Apr 2026

Selected (Researcher track) among 300+ builders to develop a neuroscience-oriented AI application using frontier multimodal models, competing for €50k+ in prizes

Microsoft x Finelogs-Biseum

AI Hackathon - Azure AI Platforms

Paris, FR

Jun 2025

Built a context-aware HR chatbot on Azure (FastAPI, CosmosDB, Azure OpenAI) with RAG over French labor law documents and role-conditioned response generation via BFS traversal of organizational hierarchies

PERSONAL INTERESTS & SKILLS

Hobbies : Running, Swimming, Reading, Piano

Languages: French (native), English (C1), Spanish (B2)

Programming Languages: Python, C/Assembly 68000, Java/C#/Scala, MATLAB, HTML/CSS/JavaScript, Oracle, SQL